

YIFAN WANG

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RESEARCH INTERESTS

Post-training and alignment of video generative models, with a focus on reinforcement learning and leveraging VLM/MLLM feedback for semantic evaluation and fine-grained credit assignment. I aim to enhance the spatiotemporal consistency and instruction-following capabilities of large-scale video generative systems.

EDUCATION

Northeastern University, Boston, MA

Sep. 2023 – Jun. 2028 (Expected)

Ph.D. in Computer Engineering under supervision of [Prof. Yun \(Raymond\) Fu](#)

Shandong University, Jinan, Shandong, P.R.China

Sep. 2018 – Jun. 2022

B.S. in Mathematics and applied mathematics

SELECTED PUBLICATIONS

- **Yifan Wang**, Gordon Guocheng Qian, Yanyu Li, Anil Kag, Yun Fu. “**Token-Level Video Reinforcement Learning**”. *Under Review*.
- **Yifan Wang**, Yanyu Li, Gordon Guocheng Qian, Sergey Tulyakov, Yun Fu, Anil Kag. “**Diffusion-DRF: Free, Rich, and Differentiable Reward for Video Diffusion Fine-Tuning**”. *NeurIPS 2026*. [\[Project page\]](#)
- **Yifan Wang**, Xu Ma, Yitian Zhang, Yizhou Wang, Zhongruo Wang, Sung Cheol Kim, Vahid Mirjalili, Visya Renganathan, Yun Fu. “GmNet: Revisiting Gating Mechanisms from A Frequency View”. *ICLR, 2026*. [\[Paper\]](#)
- **Yifan Wang**, Yun Fu. “Understanding, Abstracting and Checking: Evoking Complicated Multimodal Reasoning in LMMs”. *ACL, 2026*
- **Yifan Wang**, Lin Zhang, Ran Song, Lin Ma, Wei Zhang. “Exploiting Inter-Sample Affinity for Knowability-Aware Universal Domain Adaptation”. *International Journal of Computer Vision (IJCV), 2023*. [\[Paper\]](#)

RESEARCH EXPERIENCE

[Creative Vision@Snap Inc.](#), *Research Intern*, Santa Monica, CA

Jun. 2025 – Jul. 2026

Mentor: Anil Kag, Gordon Guocheng Qian, Yanyu Li and Sergey Tulyakov

- **Overview:** Worked on post-training and reinforcement learning for video diffusion models, developing effective and efficient methods to improve their general capabilities.
- **TVRL: Token-Level Video Reinforcement Learning**
Research Brief: Proposed TVRL, an RL framework for text-to-video diffusion that separates rollout-level reward from token-level credit assignment. A frozen VLM provides both the reward (answer log-likelihood on prompt-derived checks) and a gradient-based reward-sensitivity map that routes the GRPO update to relevant video tokens, without reward-model training or dense labels. Achieves 57.69 on VBench-2.0 (+3.2 over GRPO, +3.6 over the base model), with consistent gains across three samplers and four reward models.
- **Diffusion-DRF: Differentiable Reward Flow for Video Diffusion Fine-Tuning**
Research Brief: Developed Diffusion-DRF, a differentiable post-training framework that uses a frozen, off-the-shelf VLM as a training-free critic for video diffusion models. The method decomposes prompts into structured evaluations of text-video alignment, physical fidelity, and visual quality, combining binary judgments with free-form explanations to provide rich and localized optimization signals without reward-model training or preference annotations. Diffusion-DRF outperforms prior reward-based fine-tuning methods, including Flow-GRPO by 4.74% on VBench-2.0.

[Smile Lab@Northeastern University](#), *Ph.D student*, Boston, MA

Sep. 2023 - Now

Advisor: Prof. Yun Raymond Fu

- **Overview:** Worked on efficient vision models, multimodal reasoning with LMMs, and few-shot recognition.
- **GmNet: Revisiting Gating Mechanisms from A Frequency View.**
Research Brief: We systematically explore the effect of gating mechanisms on the training dynamics of neural networks from a frequency perspective. Leveraging insights on GLUs, we propose a Gating Mechanism Network (GmNet), a lightweight model designed to efficiently utilize the information of various frequency components. GmNet achieves impressive performance in terms of both effectiveness and efficiency in the image classification task.
- **Evoking Complicated Multimodal Reasoning in LMMs.**
Research brief: Developed UnAC (Understanding, Abstracting, and Checking), a prompting framework for improving complex multimodal reasoning in LMMs. UnAC encourages models to capture finer visual details, abstract relevant information, and verify intermediate reasoning steps..
- **Hyperbolic Few-shot Open-set Recognition.**
Research brief: Proposed a hyperbolic framework based on a Hyperbolic Conditional Variational Autoencoder to mitigate representation bias caused by data scarcity and better capture hierarchical class structures.

[Vsislab@Shandong University](#), *Research Assistant*, Jinan, Shandong, P.R.China

Mar, 2021 – Mar, 2023

Advisor: Prof. [Wei Zhang](#)

- **Exploiting Inter-Sample Affinity for Knowability-Aware Universal Domain Adaptation.**
Research Brief: Proposed a novel UniDA framework to fully exploring the inter-sample affinity between known and unknown samples. Specifically, we introduced a knowability-based labeling scheme which can be divided into two steps: (1) Knowability-guided detection of known and unknown samples based on the intrinsic structure of the neighborhoods of samples. (2) Label refinement based on neighborhood consistency to relabel the target samples.
- **Neighborhood-Aware Mutual Information Maximization for Source-Free Domain Adaptation**
Research brief: Proposed the neighborhood-aware mutual information maximization (NAMI), which maximizes the mutual information (MI) between the representations of target domain and the corresponding neighborhood. Incorporated the consistency constraint between different views into the method to encourage the model to be robust to changes in the sample space and thus further alleviate the problem of domain gap.
- **Dual-Graph Contrastive Learning for Unsupervised Person Re-Identification**
Research brief: Proposed an effective contrastive learning framework called dual-graph contrastive learning for the fully unsupervised person re-ID, which can fully utilize the intrinsic structure of the unlabeled data and better reduce the high intra-class variance. Proposed a camera-aware augmentation scheme that exploits the semantic information between samples and their corresponding camera-level centroids to relieve the domain gap between cameras.

SKILLS AND INTERESTS

- *Programming:* Python, MATLAB
- *Frameworks & Tools:* PyTorch, NumPy, scikit-learn